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Using neurocognitive phenotypes to inform interventions for adult survivors of childhood cancer

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Abstract

Background: Neurocognitive impairments are sequelae of childhood cancer treatment, however little guidance is given to clinicians on common phenotypes of impairment or modifiable risk factors that could lead to personalized interventions in survivorship.

Methods: Standardized clinical testing of neurocognitive function was conducted in 2958 (74.1%) eligible survivors, who were at least 5 years postdiagnosis and aged older than 18 years, and 477 community controls. Impairment was examined across 20 measures, and phenotypes were determined by latent class analysis. Multinomial logistic regression was used to estimate risk for phenotype, predicted by cancer diagnosis and treatment exposures, chronic health conditions, and lifestyle, adjusted for sex and age. Associations between phenotypes and social attainment were examined.

Results: Five neurocognitive phenotypes were identified in survivors (global impairment 3.7%, impaired attention 5.0%, memory impairment 7.2%, processing speed and executive function impairment 9.3%, no impairment 74.8%). Risk of global impairment was associated with severe chronic health condition burden (odds ratio [OR] = 20.17, 95% confidence interval [CI] = 11.41 to 35.63) including cerebrovascular disease (OR = 14.5, 95% CI = 5.47 to 38.44) and cerebrovascular accident (OR = 14.7, 95% CI = 7.50 to 26.40). Modifiable risk factors, such as quitting smoking, reduced risk for global impairment (OR = 0.21, 95% CI = 0.06 to 0.66). Low physical activity increased risk for global impairment (OR = 4.54, 95% CI = 2.86 to 7.21), attention impairment (OR = 2.01, 95% CI = 1.41 to 2.87), processing speed and executive function impairment (OR = 1.90, 95% CI = 1.46 to 2.48), and memory impairment (OR = 2.09, 95% CI = 1.54 to 2.82).

Conclusions: Results support the clinical utility of neurocognitive phenotyping to develop risk profiles and personalized clinical interventions, such as preventing cerebrovascular disease in anthracycline-treated survivors by preventing hypercholesterolemia, smoking, and sedentary lifestyle, to reduce the risk for global impairment.

Neurocognitive impairments are common sequelae of childhood cancer and treatment. However, variability in the specific type and magnitude of impairment is observed across diagnoses (1-3), treatment exposures (4), and physical comorbidities (5,6), with studies reporting prevalence and predictors of each individual neurocognitive ability. In fact, current cancer survivorship guidelines (www.survivorshipguidelines.org) describe potential neurocognitive late effects in terms of individual functional deficits by therapeutic risk and offer little guidance to clinicians, in regard to risk of chronic health conditions on neurocognition, beyond recommending baseline testing and neuropsychological referral, as clinically indicated by evidence of impaired educational or vocational progress (7). Additionally, the type of specific neurocognitive impairment may change or expand over time or decline later in survivorship (8).

Clinical interpretation of neurocognitive impairment typically considers the pattern of function across multiple measures to select specific treatments for the phenotype the individual may display. Multimorbidity across neurocognitive domains is common and drives individual treatment plans and interventions (9). Moreover, patient-specific factors may contribute to different neurocognitive phenotypes within and between individuals over the lifespan. For example, studies in noncancer populations have linked specific neurocognitive phenotypes to type 1 and type 2 diabetes, childhood psychiatric disorders, and Parkinson disease (10-12). Defining the most common phenotypes and predictors in cancer survivors would inform personalized interventions to address the unique needs of survivors.

The primary aim of the current study was to identify common patterns of impaired neurocognitive abilities across multiple

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assessed domains and identify predictors of these patterns from cancer diagnosis, treatment, sex, and lifestyle factors. We hypothesized that we would find discrete phenotypes of neurocognitive impairment that were mutually exclusive across diagnosis groups. We postulated that these discrete phenotypes would be associated with not only cancer treatment, such as cranial radiation, but also chronic health conditions and lifestyle choices.

Methods

Study population

Participants were enrolled in the St Jude Lifetime Cohort Study (13), a retrospectively identified cohort with prospective follow-up of 5 or more years survivors diagnosed with childhood cancer between 1962 and 2012 and treated at St Jude Children's Research Hospital. To limit variability associated with rapid developmental change during childhood, survivors were included in the current analyses if they were 18 years and older. Participants were excluded if they had a history of a noncancer-related neurodevelopmental disorder, a genetic disorder associated with neurocognitive impairment (ie, Trisomy 21), brain injury unrelated to cancer, or lacked proficiency in English. Of 3994 eligible survivors, 465 (11.6%) actively refused, 238 (6.0%) were nonresponders, 129 (3.2%) were unable to be contacted, and 204 (5.1%) had insufficient neurocognitive data for analyses with more than 2 tests missing; there were 2958 (74.1%) evaluable participants (see Figure 1). Community controls (n = 447) were recruited as a comparison group, with the same exclusion criteria as survivors in addition to having a negative history of childhood cancer (13). Controls were not first-degree relatives or friends of cancer survivors recruited from the same geographical areas as the St Jude Lifetime Cohort Study survivor cohort. All participants were provided financial support for travel, room and board, and monetary compensation while they were on campus participating in this study. The study protocol was approved by the institutional review board at St Jude Children's Research Hospital, and written informed consent was obtained from all participants.

Procedure

Participants completed a comprehensive battery of neurocognitive tests administered by certified examiners, under the general supervision of a board-certified clinical neuropsychologist, in a dedicated evaluation room using standard clinical procedures. A total of 20 neurocognitive abilities were assessed across neurocognitive domains of intelligence and academics, attention, processing and psychomotor speed, memory, and executive function, with 4 measures in each of the 5 domains (Supplementary Table 1, available online) (14). Order of testing was fixed, and participants' schedules were adjusted with scheduled breaks to limit fatigue. Examiners completed a test evaluation form that included reported sleep the night prior to testing and current fatigue, as well as other potential distractions and motivation on the day of

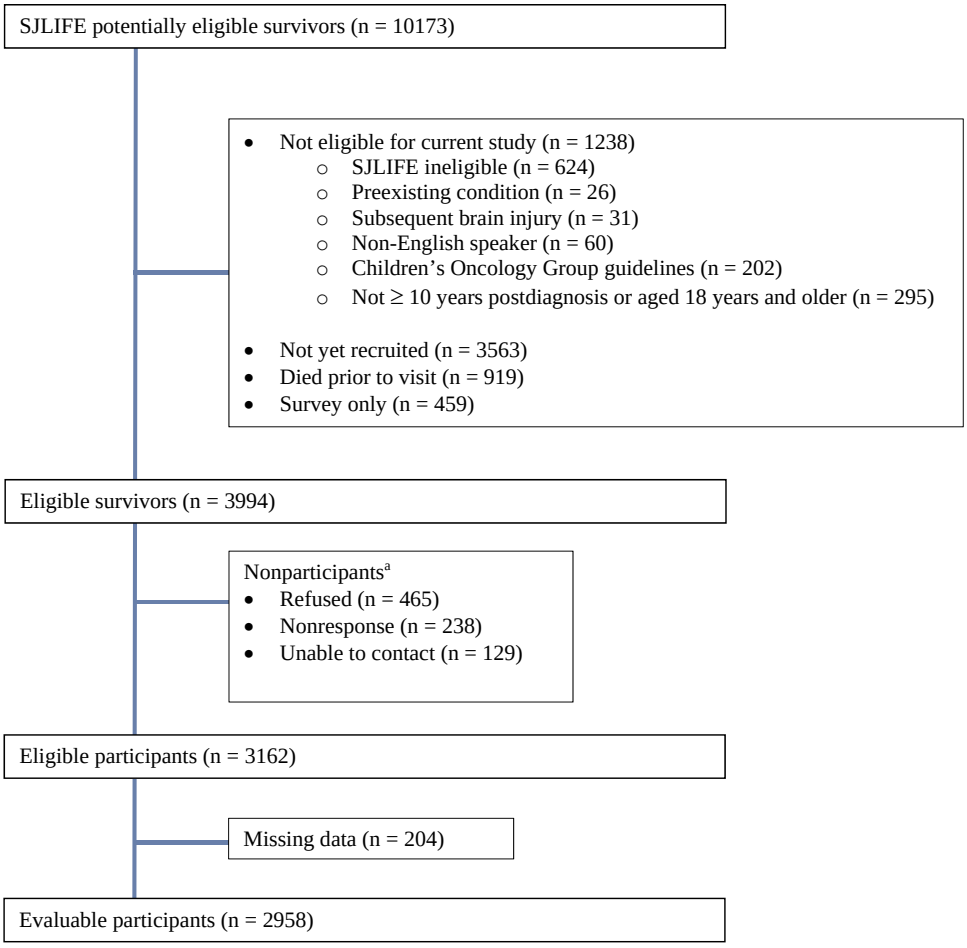

Figure 1. CONSORT diagram. [a]Comparability between nonparticipants and participants has been previously published. SJLIFE = St Jude Lifetime Cohort Study.

testing. Test performance was converted to age-adjusted z scores referenced to general population normative data, with an expected performance mean of 0 and standard deviation of 1.0. Neurocognitive outcomes were collected from 2008 to 2017 in survivors and 2014 to 2017 for controls.

Exposures included primary diagnosis and treatment, including cumulative doses of chemotherapy, dosimetry for maximum dose of cranial radiation therapy, and neurosurgical resection abstracted from medical records. Chemotherapy cumulative doses used in analyses included intrathecal methotrexate, high-dose intravenous methotrexate (defined as ≥1 g/m2 per dose), anthracyclines (doxorubicin equivalent dose), high-dose intravenous cytarabine, alkylating agents (cyclophosphamide equivalent dose), platinum agents, epipodophyllotoxins, vinca alkaloids, and corticosteroids (prednisone equivalent dose).

The presence and severity of chronic health conditions and age at onset were verified by physical exam, medical interview with a licensed health-care provider, and medical testing (echocardiogram, pulmonary function testing, electrocardiogram, and so forth) during the same multiday visit. These evaluations were used to identify chronic health conditions that occurred at or prior to neurocognitive testing. Chronic health conditions were graded for severity using the National Cancer Institute's Common Terminology Criteria for Adverse Events version 4.3 including grade 1 (mild or asymptomatic), grade 2 (moderate), grade 3 (severe or disabling), and grade 4 [life-threatening (15)] and then combined by frequency and grade into a burden score (no or low, medium, high, severe; Supplementary Table 2, available online) (6).

Sociodemographic (ie, age, sex, race, and ethnicity), smoking status, social attainment, and physical activity were collected from self-report questionnaires. Long-term socioeconomic status was assessed more than 5 years from diagnosis to measure downstream effects of modifiable risk factors and neurocognitive phenotypes on quality-of-life concerns. Years of education was not used as a covariate in this model, as education before treatment and age at diagnosis are colinear and only one is required to be present in statistical modeling. Parental education at the time of diagnosis was unavailable for most participants. Heavy drinking was defined as 6 or more drinks in 1 day for men and 5 or more drinks in 1 day for women at least once per month. Physical activity was dichotomized as meeting the Centers for Disease Control and Prevention guidelines (any combination of 150 minutes of moderate or 75 minutes of vigorous physical activity per week) and not meeting the guidelines. Smoking was defined as consuming at least 100 cigarettes and was further categorized into current or former status.

Statistical analysis

Descriptive characteristics for demographic, clinical, lifestyle, and social characteristics were calculated. Neurocognitive performance was converted to age-adjusted z scores using population normative data, and performance was compared between survivors and controls using general linear models. Mild to severe neurocognitive impairment was defined as a z score of no more than -1.0 for each of the 20 neurocognitive abilities, equivalent to approximately the 16th percentile in the general population, and serious, moderate to severe neurocognitive impairment was conservatively defined as a z score of no more than -2.0 for each of the abilities, equivalent to the second percentile in the general population. Neurocognitive phenotypes were identified using latent class analysis on the 20 neurocognitive abilities assessed and were dichotomized (yes or no) by presence of serious impairment, with the number of classes determined by quantitative model fit statistics (Bayesian information criterion, sample-size adjusted Bayesian information criterion, entropy, Vuong–Lo–Mendell–Rubin Likelihood Ratio test), and qualitative assessment (ie, most clinically parsimonious model). Multinomial logistic regression was used to estimate the odds of displaying a phenotype associated with demographic, treatment characteristics, lifestyle behaviors (ie, smoking, alcohol use, and physical activity), and chronic health conditions. Separate models were generated for chronic health condition burden, grade 2-4 chronic health condition for organ systems, and grade 2-4 chronic health condition on the individual condition level if at least 10% frequency in the sample or potential of high clinical impact on the brain. Models were adjusted for sex, age at diagnosis, and time since diagnosis, and the cutoff for statistical significance was set at a P value less than .05. All analyses were conducted in SAS 9.4 (SAS Institute, Cary, NC, USA) and Mplus 8.0 (Muthén & Muthén, Los Angeles, CA, USA).

Results

Demographic and clinical characteristics are presented in Table 1. There were more female (55.3% vs 47.4%), White Hispanic (3.1% vs 1.6%), and White non-Hispanic (85.7% vs 82.4%) controls than survivors. This is consistent with previous studies that have demonstrated fewer male and minority group volunteers (13). Among the 2958 survivors (age at evaluation [median = 33.0, interquartile range (IQR) = 26.7-40.5 years], time from diagnosis [median = 24.7, IQR = 18.2-32.4 years]), the most frequent primary cancer diagnoses were acute lymphoblastic leukemia (ALL; 33.6%), central nervous system tumor (12.3%), and Hodgkin lymphoma (11.2%). Compared with controls, survivors demonstrated lower performance on all 20 neurocognitive measures (all P <.01; Supplementary Table 3, available online).

Neurocognitive phenotypes

Of the survivors, 25.2% demonstrated an impaired neurocognitive phenotype. Global impairment was identified in 3.7% (n = 110) survivors, isolated attention impairment in 5.0% (n = 149), isolated memory impairment in 7.2% (n = 212), and combined processing speed and executive function impairment in 9.3% (n = 274), and the remaining 74.8% (n = 2213) of survivors demonstrated no serious neurocognitive impairment (Figure 2; Supplementary Table 4, available online) (16,17). The percentage of survivors in each neurocognitive phenotype by diagnostic group is shown in Figure 3. Within groups, survivors diagnosed with central nervous system tumors demonstrated a higher percentage of global impairment, whereas no survivors of Ewing sarcoma, Hodgkin lymphoma, or soft tissue sarcoma demonstrated global impairment. Compared with males, females had a lower risk of processing speed and executive function impairment (odds ratio [OR] = 0.60, 95% confidence interval [CI] = 0.45 to 0.79) but a higher risk of memory impairment (OR = 2.07, 95% CI = 1.52 to 2.83).

Treatment associations by phenotype

Survivors treated with neurosurgery or cranial radiation were more likely to demonstrate global impairment (neurosurgery: OR = 3.61, 95% CI = 1.74 to 7.50; cranial radiation: OR = 1.47 per 10 Gy, 95% CI = 1.29 to 1.68), processing speed and executive function impairment (neurosurgery: OR = 1.71, 95% CI = 1.05 to 2.77; cranial radiation: OR = 1.32 per 10 Gy, 95% CI = 1.21 to 1.44), or memory impairment (neurosurgery: OR = 1.68, 95% CI = 1.01

Table 1. Demographic, clinical, and treatment characteristics[a]

Characteristics	Survivors (n = 2958)	Controls (n = 447)	P
Demographic			
Age at evaluation, median (IQR), y	33.0 (26.7-40.5)	34.1 (27.5-41.7)	.08
Age at diagnosis, median (IQR), y	7.3 (3.4-13.2)		
Time since diagnosis, median (IQR), y	24.7 (18.2-32.4)		
Sex, No. (%)			
Male	1557 (52.6)	200 (44.7)	.002
Female	1401 (47.4)	247 (55.3)	
Ethnicity, No. (%)			
Hispanic	46 (1.6)	14 (3.1)	.02
Non-Hispanic	2912 (98.4)	433 (96.9)	
Race, No. (%)			
Black, non-Hispanic	442 (14.9)	34 (7.6)	<.0001
Other (Amerindian, Asian, Pacific Islanders)	80 (2.7)	30 (6.7)	
White, non-Hispanic	2436 (82.4)	383 (85.7)	
Education, No. (%)			
College graduate or above	1025 (37.3)	231 (55.0)	<.0001
Below college graduate	1723 (62.7)	189 (45.0)	
Employment, No. (%)			
Full-time	1853 (67.7)	327 (76.4)	<.0001
Not full-time	884 (32.3)	101 (23.6)	
Marital status, No. (%)			
Single	1058 (36.7)	98 (22.3)	<.0001
Ever married	1826 (63.3)	341 (77.7)	
Personal income, No. (%)			
<$40 000	2087 (73.9)	253 (58.6)	<.0001
$40 000-$59 999	364 (12.9)	84 (19.4)	
$60 000-$79 999	168 (5.9)	46 (10.6)	
$80 000-$99 999	79 (2.8)	26 (6.0)	
≥$100 000	126 (4.5)	23 (5.3)	
Independent living, No. (%)			
Yes	1873 (65.0)	340 (77.4)	<.0001
No	1009 (35.0)	99 (22.6)	
Requires assistance with personal or routine needs, No. (%)			
Yes	273 (9.5)	12 (2.7)	<.0001
No	2608 (90.5)	425 (97.3)	
Treatment, median (IQR)[b]			
Cranial radiation dose, Gy (n = 830)	24.0 (18.0-40.0)		
Intrathecal methotrexate, mg (n = 1220)	144.0 (84.0-180.0)		
High-dose intravenous methotrexate, mg/m^2 (n = 851)	16114.0 (4460.0-22 034.4)		
Anthracyclines,[c] mg/m^2 (n = 1725)	168.7 (102.6-248.3)		
High-dose intravenous cytarabine, mg/m^2 (n= 210)	17634.0 (8000.0-21 315.1)		
Alkylating agents,[d] mg/m^2 (n = 1758)	8629.0 (5214.8-12 918.0)		
Heavy metals, mg/m^2 (n = 375)	553.3 (354.4-2722.7)		
Epipodophyllotoxins, mg/m^2 (n = 1122)	2751.4 (1200.0-12 441.0)		
Vinca alkaloids, mg/m^2 (n= 2080)	37.2 (10.3-55.8)		
Corticosteroids,[e] mg/m^2 (n = 1353)	3920.0 (1500.0-10 300.0)		
Neurosurgery (n = 401)	N/A		
Primary diagnosis, No. (%)			
Acute lymphoblastic leukemia	994 (33.6)		
Acute myeloid leukemia	102 (3.4)		
Central nervous system tumor	363 (12.3)		
Ewing sarcoma	77 (2.6)		
Germ cell tumor	63 (2.1)		
Hodgkin lymphoma	331 (11.2)		
Neuroblastoma	114 (3.9)		
Non-Hodgkin lymphoma	215 (7.3)		
Osteosarcoma	119 (4.0)		
Retinoblastoma	93 (3.1)		
Rhabdomyosarcoma	96 (3.2)		
Soft tissue sarcoma	68 (2.3)		
Wilms tumor	179 (6.1)		
Other[f]	144 (4.9)		
Chronic health condition severity or burden, No. (%)			
No or low	209 (7.1)		
Medium	1455 (49.2)		
High	994 (33.6)		
Severe	300 (10.1)		

[a] Total numbers for each category vary because of missing data. IQR = interquartile range; N/A = not applicable.
[b] For treatment variables, refers to cumulative median dose within 10 years of primary cancer diagnosis among those who received the exposure.
[c] Other category includes hepatoblastoma, melanoma, carcinomas, other leukemias, myelodysplastic syndrome, peripheral neuroepithelioma, and other neoplasms.
[d] Doxorubicin equivalent dose.
[e] Cyclophosphamide equivalent dose.
[f] Prednisone equivalent dose.

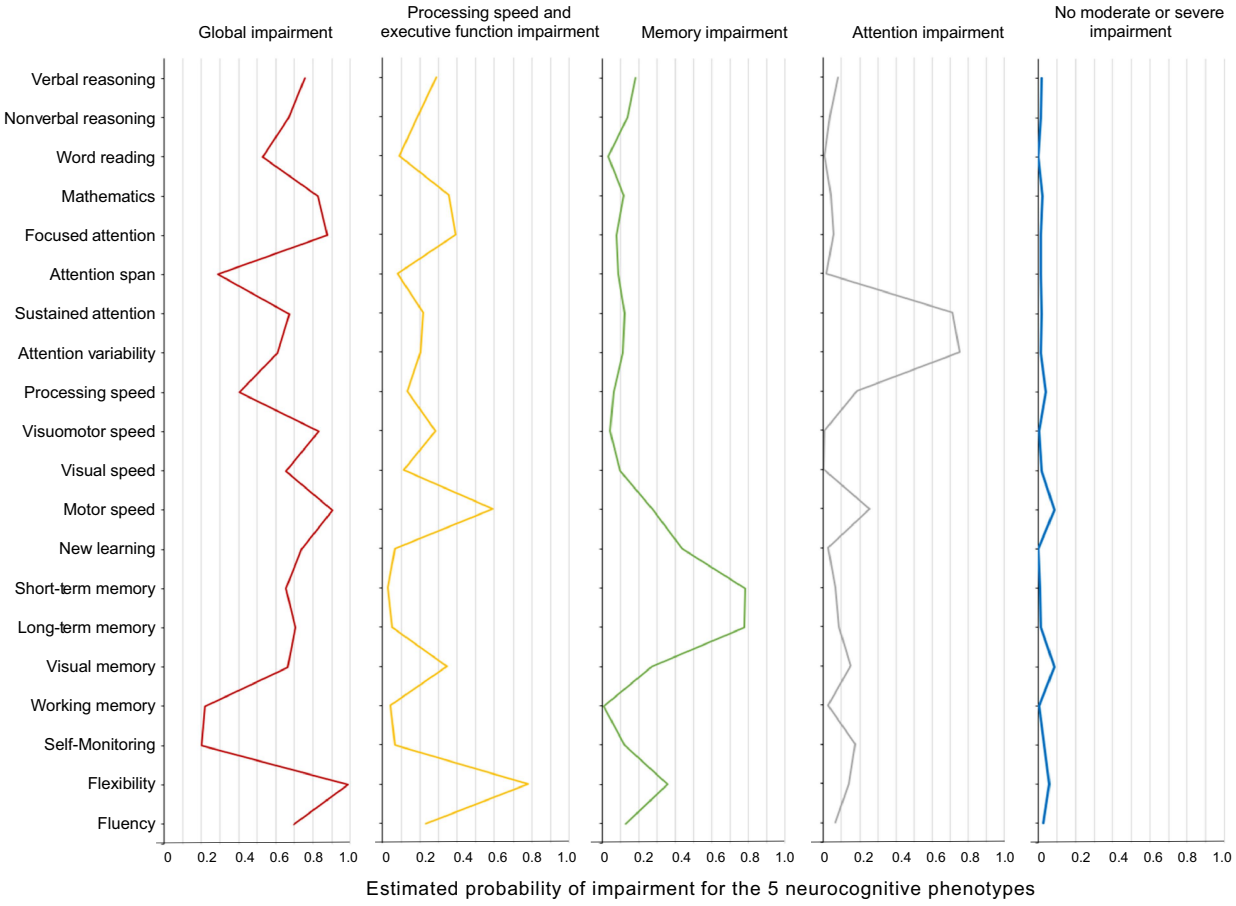

Figure 2. Probability of impairment on each neurocognitive ability for the 5 phenotypes in survivors.

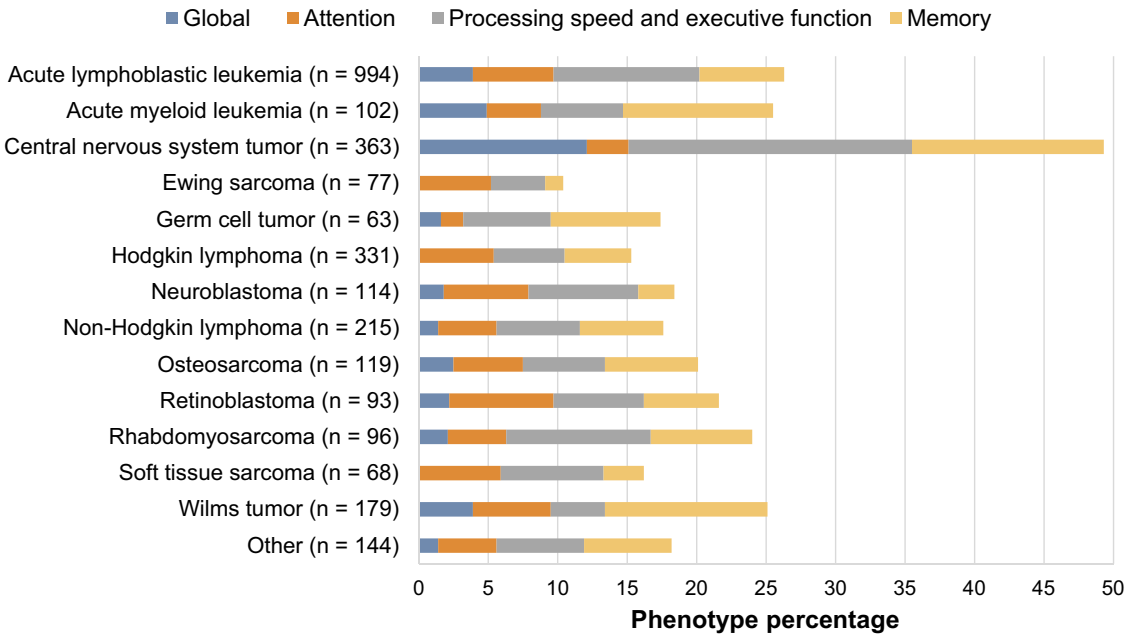

Figure 3. Neurocognitive phenotype by primary diagnosis. Percentage of primary diagnosis subgroup in each neurocognitive impairment phenotype (no impairment group is not shown and would bring totals to 100%). The "other" category includes hepatoblastoma, melanoma, carcinomas, other leukemias, myelodysplastic syndrome, peripheral neuroepithelioma, and other neoplasms.

to 2.79; cranial radiation: OR = 1.11 per 10 Gy, 95% CI = 1.00 to 1.23) compared with no severe impairment (Table 2). Anthracyclines were associated with elevated odds of attention impairment (OR = 1.15 per 100 mg/m^2, 95% CI = 1.003 to 1.32) compared with no impairment. Hydrocephalus requiring shunting carried the highest risk for global impairment (OR = 19.21, 95% CI = 11.03 to 33.46).

Chronic health conditions by phenotype

Risk of global impairment was 20-fold greater for survivors with severe chronic health conditions burden compared with survivors with no or low or medium burden (OR = 20.17, 95% CI = 11.41 to 35.63). High chronic health conditions were also associated with global impairment (OR = 5.01, 95% CI = 3.01 to 8.35), processing speed and executive function impairment (OR = 2.13, 95% CI = 1.60 to 2.84), and memory impairment (OR = 1.64, 95% CI = 1.19 to 2.24) compared with survivors with no or low or medium burden (Table 3). Frequency of grade 2-4 vs 0-1 chronic health conditions (Supplementary Table 5, available online) and associations with neurocognitive phenotype by organ systems for all 12 systems were calculated (Supplementary Tables 6 and 7, available online). All organ systems except immunology and renal and urinary showed statistically significant associations with global impairment (eg, ORs range from 1.96 [95% CI = 1.31 to 2.93] for gastrointestinal system to 10.41 [95% CI = 6.49 to 16.69] for neurology), and all systems except hematology, immunology/infectious, and renal and urinary were associated with processing speed and executive function impairment (ORs range from 1.36 [95% CI = 1.05 to 1.77] for musculoskeletal system to 4.41 [95% CI = 3.38 to 5.77] for neurology). Neurology and pulmonary systems were associated with isolated attention (neurology: OR = 1.94, 95% CI = 1.38 to 2.74; pulmonary: OR = 2.40, 95% CI = 1.70 to 3.38) and isolated memory impairment (neurology: OR = 2.18, 95% CI = 1.63 to 2.92; pulmonary: OR = 1.65, 95% CI = 1.22 to 2.24).

For individual chronic health conditions, peripheral sensory and motor neuropathy increased odds of impairment in all phenotypes (global impairment: OR = 10.99, 95% CI = 6.92 to 17.44, and OR = 8.29, 95% CI = 11.59 to 28.84; attention impairment: OR = 3.02, 95% CI = 1.86 to 4.90, and OR = 3.39, 95% CI = 2.07 to 5.57; processing speed and executive impairment: OR = 3.68, 95% CI = 2.53 to 5.35, and OR = 8.13, 95% CI = 5.78 to 11.43; memory impairment: OR = 2.47, 95% CI = 1.59 to 3.83, and OR = 3.63, 95% CI = 2.37 to 5.54, respectively; Figure 4; Supplementary Table 6, available online). Several phenotypes were associated with endocrine conditions: adrenal insufficiency (ORs range from 12.03 [95% CI = 6.70 to 21.62] for global impairment to OR = 2.91 [95% CI = 1.54 to 5.51] for memory impairment) and child and adult growth hormone deficiency (ORs range from 6.83 [95% CI = 4.24 to 11.02] for global impairment to OR = 1.91 [95% CI = 1.11 to 3.27] for memory impairment). Phenotypes were also associated with pulmonary diffusion deficits (ORs ranging from 2.64 [95% CI = 1.83 to 3.81] for attention impairment to OR = 1.82 [95% CI = 1.15 to 2.90] for global impairment) and restrictive pulmonary deficits (ORs ranging from 2.87 [95% CI = 1.67 to 4.94] for global impairment to OR = 1.98 [95% CI = 1.17 to 3.35] for attention impairment). Anemia carried the highest odds of attention impairment (OR = 3.78, 95% CI = 1.38 to 10.36).

Modifiable risk factors by phenotype

Modifiable lifestyle behaviors carried additional risks for survivors (Table 4; Supplementary Table 7, available online). Being physically inactive increased a survivor's risk for global impairment (OR = 4.54, 95% CI = 2.86 to 7.21), attention impairment (OR = 2.01, 95% CI = 1.41 to 2.87), processing speed and executive function impairment (OR = 1.90, 95% CI = 1.46 to 2.48), and memory impairment (OR = 2.09, 95% CI = 1.54 to 2.82). Current smokers carried a higher risk for attention impairment (OR = 2.32, 95% CI = 1.59 to 3.39), processing speed and executive function impairment (OR = 1.39, 95% CI = 1.02 to 1.89), and memory impairment (OR = 1.87, 95% CI = 1.35 to 2.60) than survivors who never smoked. Quitting smoking reduced the risk for global impairment (OR = 0.21, 95% CI = 0.06 to 0.66).

Phenotype prediction of social attainment

Survivors had lower incomes, were less educated, married, and employed full-time; fewer lived alone; and more required assistance with personal needs (Table 1; all $P < .0001$). Membership in any impaired phenotype category was associated with elevated risk of poor social attainment (Supplementary Table 8, available online). The risk for requiring assistance with routine or personal needs varied from 2.67 (95% CI = 1.71 to 4.18) for those with attention-only impairments to 11.80 (95% CI = 9.08 to 15.34) for those with global impairment. Similar patterns were seen for independent living, education attainment, employment, and personal income. Only attention impairment carried no risk for never being married.

Discussion

Using nearly 3000 adult survivors of childhood cancer, we discovered 5 distinct phenotypic patterns of neurocognitive impairment that mimic clinical presentations. Our study differs from previous studies in 2 important ways. First, prior research of neurocognitive outcomes across differing diagnosis groups required harmonization of outcomes from a heterogenous mix of neurocognitive tests or self-report questionnaires that varied by cancer diagnosis and/or institution. Our study used the same standard neuropsychological assessments on all cancer diagnosis groups starting 5 years after diagnosis along with concurrent and equivalent medical and physical evaluation. Secondly, previous reports focus on reporting associations between demographic, treatment, and chronic health conditions with individual neuropsychological tests, which may not adequately describe the phenotypes of neurocognitive impairment typically seen in individual survivors. We show that survivors with similar diagnoses and treatments can demonstrate distinct phenotypes of impairment and that these phenotypes are uniquely associated with risk factors, such as sex, chronic health conditions, and modifiable life-style choices. This extends the traditional approach of examining individual test measures and informs clinicians about specific patterns of neurocognitive comorbidity and severity. Given these patterns, the selection of specific interventions to target neurocognitive impairment should be based on the distinct neurocognitive phenotype, sex, prevalent chronic health conditions, and modifiable risk factors rather than solely on cancer diagnosis, treatment exposure, or individual neurocognitive test performance.

One of the novel findings of this study was that isolated attention impairment was associated with higher cumulative dose of anthracyclines. Although the mechanism behind this finding may be through anthracycline-induced cardiac dysfunction (18), only hypercholesterolemia was associated with the isolated attention impairment phenotype. This suggests that early interventions to aggressively prevent or treat hypercholesterolemia

Table 2. Multinomial logistic regression results of demographic, treatment, and chronic health predictors of neurocognitive phenotype class membership

	Global impairment OR (95% CI)[a]	Attention impairment OR (95% CI)[a]	Processing speed and executive function impairment OR (95% CI)[a]	Memory impairment OR (95% CI)[a]
Model 1: treatment				
Female, referent male	1.35 (0.87 to 2.10)	1.32 (0.93 to 1.88)	0.60 (0.45 to 0.79)	2.07 (1.52 to 2.83)
Age at diagnosis, per 1 year older	0.85 (0.80 to 0.90)	0.95 (0.92 to 0.98)	0.97 (0.95 to 1.00)	0.99 (0.96 to 1.02)
Time since diagnosis, per 1 year	1.01 (0.98 to 1.04)	1.00 (0.98 to 1.02)	1.00 (0.98 to 1.02)	1.00 (0.98 to 1.02)
Neurosurgery, yes vs no	3.61 (1.74 to 7.50)	0.98 (0.47 to 2.04)	1.71 (1.05 to 2.77)	1.68 (1.01 to 2.79)
Cranial radiation dose, per 10 Gy	1.47 (1.29 to 1.68)	1.14 (0.99 to 1.30)	1.32 (1.21 to 1.44)	1.11 (1.01 to 1.23)
Intrathecal methotrexate, per 100 mg	1.29 (0.93 to 1.78)	1.06 (0.81 to 1.39)	1.10 (0.89 to 1.35)	0.76 (0.56 to 1.03)
High-dose intravenous methotrexate, per 1 g/m^2	1.00 (0.98 to 1.02)	1.00 (0.99 to 1.01)	1.00 (0.99 to 1.01)	1.00 (0.99 to 1.01)
Anthracyclines,[b] per 100 mg/m^2	1.02 (0.83 to 1.25)	1.15 (1.003 to 1.32)	0.95 (0.83 to 1.08)	0.90 (0.78 to 1.04)
High-dose intravenous cytarabine, per 1 g/m^2	1.00 (0.94 to 1.06)	1.01 (0.98 to 1.04)	0.98 (0.95 to 1.02)	1.03 (1.00 to 1.05)
Alkylating agents,[c] per 100 mg/m^2	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.00)	1.00 (0.99 to 1.00)
Heavy metals, per 100 mg/m^2	1.02 (1.00 to 1.04)	0.99 (0.96 to 1.02)	1.02 (1.00 to 1.03)	1.01 (0.99 to 1.03)
Epipodophyllotoxins, per 100 mg/m^2	0.99 (0.98 to 1.00)	1.00 (0.99 to 1.00)	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.01)
Vinca alkaloids, per 100 mg/m^2	0.63 (0.21 to 1.92)	0.89 (0.48 to 1.64)	0.98 (0.77 to 1.23)	1.03 (0.91 to 1.17)
Corticosteroids,[d] per 100 mg/m^2	1.01 (1.00 to 1.01)	1.00 (0.99 to 1.01)	1.00 (0.99 to 1.00)	1.00 (0.99 to 1.00)
Model 2: chronic health conditions				
High chronic health condition burden, referent grade none or 1	5.01 (3.01 to 8.35)	1.26 (0.86 to 1.85)	2.13 (1.60 to 2.84)	1.64 (1.19 to 2.24)
Severe chronic health condition burden, referent grade none or 1	20.17 (11.41 to 35.63)	2.53 (1.50 to 4.25)	3.83 (2.55 to 5.74)	2.19 (1.34 to 3.56)

[a] For all analyses, impairment classes were compared with no severe impairment. In model 2, high chronic health condition burden and severe chronic health condition burden were compared with no or low or moderate chronic health condition burden; no/low and moderate chronic health condition burden were combined because of low frequencies across the impairment classes. Model 2 was also adjusted for sex, age at diagnosis, and time since diagnosis. CI = confidence interval; OR = odds ratio.
[b] Doxorubicin equivalent dose.
[c] Cyclophosphamide equivalent dose.
[d] Prednisone equivalent dose.

Table 3. Multinomial logistic regression results of grade none or 1 vs grade 2-4 chronic health conditions at the organ system level predicting neurocognitive phenotype class membership

Organ system, referent grade none or 1	Global impairment OR (95% CI)[a]	Attention impairment OR (95% CI)[a]	Processing speed and executive function impairment OR (95% CI)[a]	Memory impairment OR (95% CI)[a]
Ocular	3.58 (2.19 to 5.84)	1.22 (0.67 to 2.22)	1.80 (1.19 to 2.72)	1.25 (0.74 to 2.13)
Cardiovascular	2.57 (1.69 to 3.91)	1.20 (0.83 to 1.72)	1.38 (1.05 to 1.82)	1.12 (0.82 to 1.53)
Auditory	5.04 (3.33 to 7.64)	1.06 (0.64 to 1.74)	2.10 (1.54 to 2.85)	1.19 (0.78 to 1.82)
Endocrine	2.77 (1.46 to 5.26)	1.45 (0.91 to 2.30)	1.90 (1.30 to 2.78)	1.13 (0.78 to 1.62)
Gastrointestinal	1.96 (1.31 to 2.93)	1.08 (0.75 to 1.55)	1.49 (1.14 to 1.94)	1.01 (0.74 to 1.38)
Hematology	3.67 (1.48 to 9.10)	2.31 (0.95 to 5.59)	1.57 (0.65 to 3.78)	3.32 (1.72 to 6.38)
Immunology, infectious	1.46 (0.92 to 2.32)	0.89 (0.56 to 1.40)	0.96 (0.68 to 1.36)	1.04 (0.71 to 1.52)
Musculoskeletal	2.54 (1.72 to 3.75)	0.94 (0.65 to 1.35)	1.36 (1.05 to 1.77)	1.09 (0.80 to 1.47)
Neurology	10.41 (6.49 to 16.69)	1.94 (1.38 to 2.74)	4.41 (3.38 to 5.77)	2.18 (1.63 to 2.92)
Pulmonary	2.37 (1.59 to 3.54)	2.40 (1.70 to 3.38)	2.04 (1.56 to 2.66)	1.65 (1.22 to 2.24)
Renal and urinary	1.22 (0.61 to 2.41)	1.58 (0.94 to 2.65)	1.09 (0.69 to 1.73)	1.24 (0.76 to 2.01)
Sexual and reproductive	2.28 (1.50 to 3.46)	1.10 (0.75 to 1.60)	1.55 (1.17 to 2.06)	1.18 (0.86 to 1.62)

[a] For all analyses, impairment classes were compared with no severe impairment. Chronic health conditions were graded using the modified version of the National Cancer Institute's Common Terminology Criteria for Adverse Events version 4.3 (grade 1, mild; grade 2, moderate; grade 3, severe or disabling; grade 4, life-threatening) (30). CI = confidence interval; OR = odds ratio.

may be needed to reduce the risk for attention impairment in those exposed to anthracyclines.

We also found that compared with males, females had a higher risk for the isolated memory impairment phenotype. In the general aging population, females are more likely than same-aged males to be diagnosed with Alzheimer disease or other forms of dementia (19). Premature aging has been demonstrated in survivors of childhood cancer (20), with higher prevalence of prefrailty and baseline frailty in females (21). Female cancer survivors are at elevated risk for reduction in estrogen levels, which typically occurs in the general population during later adulthood (22) and increases susceptibility to depression (23) and comorbid memory problems (24).

The risk of global impairment was 20 times higher for survivors with severe chronic health condition burden compared with survivors with no or low or medium burden. Several associations were identified between neurocognitive phenotypes and grade 2-4 chronic health conditions at the organ and the individual condition level. It is important to acknowledge that associations do not necessarily imply causation. For example, the associations between peripheral motor and sensory neuropathies and all 4 phenotypes of neurocognitive impairment are likely due to the

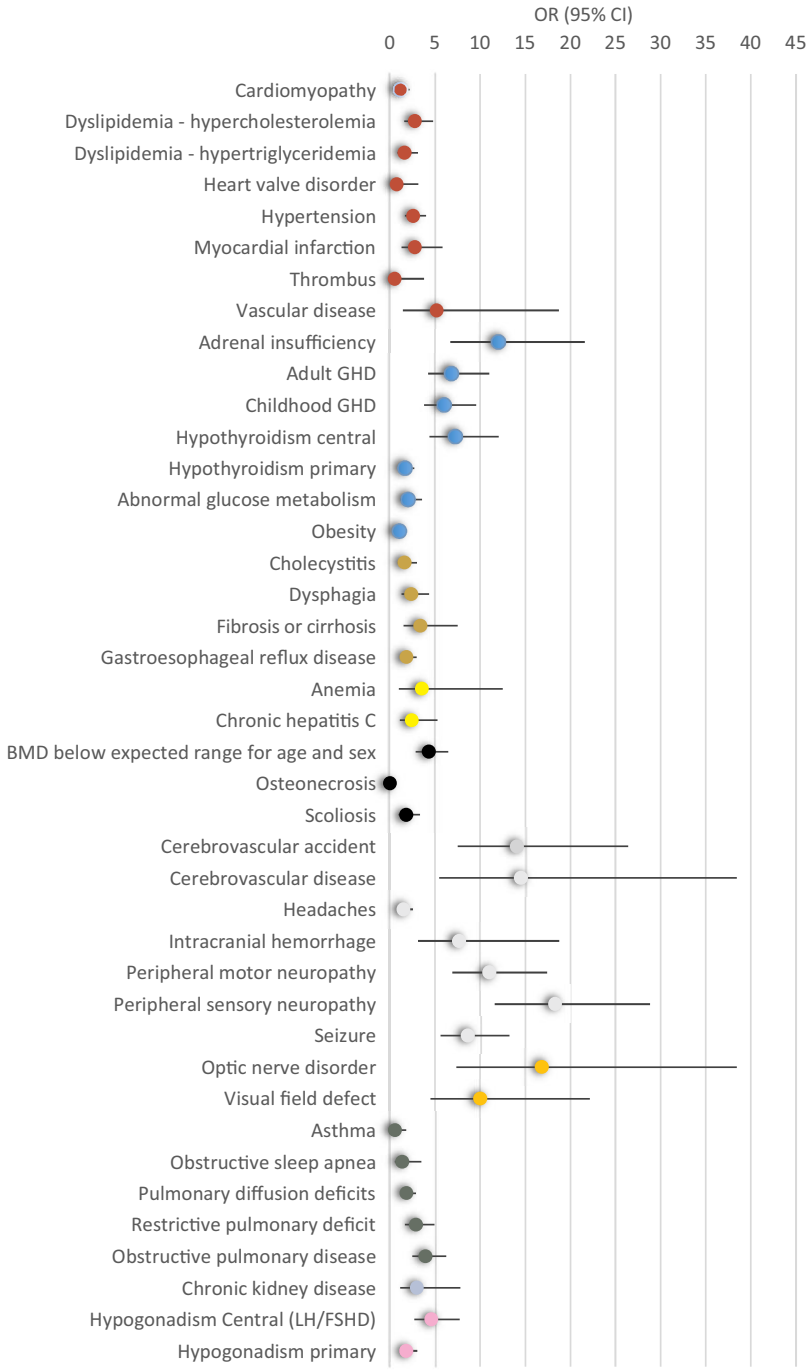

Figure 4. Multinomial logistic regression results of grade none-1 vs grade 2-4 chronic health conditions at the organ system level predicting global impairment phenotype. CI = confidence interval; FSHD = follicle-stimulating hormone deficiency; GHD = growth hormone deficiency; LH = luteinizing hormone; OR = odds ratio.

Table 4. Multinomial logistic regression results of lifestyle behaviors predicting neurocognitive phenotype[a]

Variable	Category	Global impairment OR (95% CI)	Attention impairment OR (95% CI)	Processing speed and executive function impairment OR (95% CI)	Memory impairment OR (95% CI)
Smoker	Former smoke vs never smoke	0.21 (0.06 to 0.66)	0.97 (0.54 to 1.75)	0.68 (0.43 to 1.07)	0.79 (0.46 to 1.34)
	Current smoke vs never smoke	0.85 (0.51 to 1.42)	2.32 (1.59 to 3.39)	1.39 (1.02 to 1.89)	1.87 (1.35 to 2.60)
Heavy drinking	Yes vs no	0.39 (0.12 to 1.25)	2.02 (1.21 to 3.39)	1.18 (0.76 to 1.85)	1.19 (0.68 to 2.07)
Physical activity	Physically inactive (<150) vs physically active (≥150)	4.54 (2.86 to 7.21)	2.01 (1.41 to 2.87)	1.90 (1.46 to 2.48)	2.09 (1.54 to 2.82)

[a] Heavy drinking was defined as 6 or more drinks in 1 day for men and 5 or more drinks in 1 day for women at least once per month. Each variable in different individual models adjusted for sex, age at diagnosis, and time since diagnosis. CI = confidence interval; OR = odds ratio.

underlying etiologies of neuropathy, such as diabetes or thyroid disease, which are known to be associated with neurocognitive impairments in noncancer populations. Growth hormone deficiency–related impairment phenotypes are likely due to growth hormone deficiency being a marker for increased cranial radiation, as found in a previous study of growth hormone deficiency in childhood survivors of ALL (25). In other examples, it is unknown whether the relationship is causational, such as whether cognitive difficulties in anemic patients are due specifically to the lack of hemoglobin, the resulting iron deficiency (26), or the development of specific cardiovascular conditions for which anemic patients are at risk (27). However, chronic health conditions have been associated with higher prevalence of late-onset neurocognitive impairment in patients with no impairment at completion of therapy (8,28).

Our results can be particularly useful to providers. Using this classification system, clinicians can effectively evaluate risk factors and impairment phenotypes to develop individualized interventions. For example, a clinician could expect a pharmacological intervention with methylphenidate may be best used in the isolated attention phenotype, while those with global impairments, who also display attention problems, would benefit from a broader strategy that included rehabilitation programs and cognitive training in combination with pharmaceuticals. Most importantly, our data suggest that clinical interventions based around physical activity and smoking cessation and prevention would benefit survivors across a broad spectrum of phenotypes. Previous studies have demonstrated a strong association between exercise intolerance in ALL survivors (29). The data from this study suggest this is generalizable to all childhood cancer survivors. Finally, and most strikingly, this data supports starting surveillance and interventions for chronic health conditions and modifiable risk factors at a much earlier age than those recommended for the general population.

The findings of our study should be interpreted in the context of its limitations. A cross-sectional study design was used, which does not allow us to examine how neurocognitive phenotypes may change over time within survivors. Also, a comprehensive set of covariates was considered in this study including demographics, treatment exposures, and chronic health conditions, however, there may be additional variables that moderate or mediate the impact of predictors on neurocognitive outcomes over time. For example, sleep quality and sleep difficulties, which are known to impact neurocognition in cancer survivors, were not consistently obtained for this study. Longitudinal study of such variables is warranted. In key demographic characteristics, our survivor population did differ from community controls by sex and race, with a slightly higher proportion of males among survivors than controls and a larger proportion of African Americans among survivors than controls. But neither sex nor race has been demonstrated to be associated with normal neurocognitive development. Years of educational attainment up to treatment was not used as a covariate in our statistical modeling as the children were diagnosed as young as 3 years of age and educational attainment was colinear with age at diagnosis. Therefore, outcomes, more than 20 years from diagnosis, such as education, employment, marital status, personal income, and independence, most likely reflect downstream effects of neurocognitive status of survivors though some, such as education, have bidirectional interactions with neurocognitive status.

The results of this study, showing that 5 neurocognitive impairment phenotypes are conserved across different diagnosis groups and treatment eras, emphasizes that health behaviors and chronic health conditions carry a greater burden for childhood cancer survivors than treatment alone. This emphasizes that survivorship care should focus on reducing this burden, which is greatest in those aged 20-39 years, in the nearly 500 000 childhood cancer survivors, who will not benefit from the changes we have made in our current therapies. The impact of these phenotypes on social attainment measures is quite clear from our data. The financial drain on survivors and families with these impairment phenotypes could be substantial as the risk for requiring assistance with routine or personal needs and not being able to live independently was statistically significantly associated with membership to any phenotype group. Thus, our results support the importance and clinical utility of neurocognitive phenotyping to develop risk profiles and clinical interventions.

Data availability

All data collected for this study are available to researchers outside of St Jude to facilitate the development of treatments and cures for childhood cancer and its long-term effects. The data that support the findings of this study are available at https://www.stjude.cloud/research-domains/cancer-survivorship. Data specific for this paper will be uploaded to https://zenodo.org concomitant with publication of the manuscript.

Author contributions

Pia Banerjee, PhD (Conceptualization; Investigation; Methodology; Validation; Visualization; Writing—original draft), Nicholas Phillips, MD, PhD (Investigation; Methodology; Validation; Visualization; Writing—original draft; Writing—review & editing),Wei Liu, PhD (Data curation; Formal analysis), Matthew J. Ehrhardt, MD (Writing—review & editing),Nickhill Bhakta, MD (Writing—review & editing),Tara Brinkman, PhD (Writing—review & editing),AnnaLynn Williams, PhD (Investigation; Methodology; Writing—review & editing),Yutaka Yasui, PhD (Investigation; Methodology; Writing—review & editing),Raja Khan, MD (Writing—review & editing),Deokumar Srivastava, PhD (Formal analysis; Supervision; Validation; Visualization; Writing—review & editing),Kirsten Ness, PhD (Writing—review & editing),Leslie Robison, PhD (Funding acquisition; Resources; Writing—review & editing),Melissa Hudson, MD (Funding acquisition; Resources; Writing—review & editing), and Kevin Krull, PhD (Conceptualization; Investigation; Methodology; Supervision; Writing—review & editing).

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Conflicts of interest

All authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Dr Banerjee reported employment and equity in Neuroglee Therapeutics. No other authors reported disclosures.

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